MoCHII, M^3, and MAPPINGS V 5.2.1: An Algorithmic and Physical Comparison

Abstract

This memo compares MOCHII (MOnte Carlo for H II regions), the public M^3 (Messenger Monte Carlo MAPPINGS V) tree at `~/RT_Codes/MAPPINGS/M3-main`, and the independent public MAPPINGS V 5.2.1 release at `~/RT_Codes/MAPPINGS/mappings_V-521`. The comparison is based on connected execution paths and quantitative tests rather than on routine names alone. MOCHII is strongest as a modern three-dimensional transport and synthetic-observation engine: octree AMR, an analytic absorption estimator, a dedicated Cartesian DDA backend, solution-driven ionization-front refinement, dust absorption and scattering, stochastic dust/PAH emission, and observer peel-off imaging. MAPPINGS V 5.2.1 is strongest as a deterministic one-dimensional general plasma engine: a 30-element option, density-dependent multilevel cooling, finite-source-age photoionization, non-equilibrium cooling, radiation pressure, and steady radiative MHD shocks with an iterated precursor. Its 1D photoionization path also connects detailed grain/PAH physics to a stochastic infrared solver. M^3 occupies a distinct middle ground: it grafts three-dimensional Monte Carlo transport onto an older MAPPINGS V 5.1.13-era physics tree and therefore combines broad MAPPINGS microphysics with arbitrary Cartesian geometry, but it does not inherit all later 5.2.1 changes or expose all parent modes through its public 3D driver. The principal limitations are correspondingly different. MOCHII presently uses low-density-limit metal cooling fits in its hot thermal loop and has an H II-region-centered ion inventory; M^3 has no AMR and a communication-heavy uniform-grid packet algorithm; and MAPPINGS V 5.2.1 is restricted to spherical or plane-parallel 1D structures, uses an outward-only diffuse-field approximation, and retains a legacy interactive global-state architecture. The recommended strategy is to retain the MOCHII transport architecture, use MAPPINGS V 5.2.1 as the current microphysics and 1D shock/time-dependence reference, and use M^3 specifically to study how MAPPINGS physics behaves after being mapped into three dimensions.

Contents

1. Purpose, scope, and basis of the comparison

MoCHII and M^3 are self-consistent three-dimensional Monte Carlo photoionization codes, whereas MAPPINGS V 5.2.1 is a deterministic one-dimensional plasma and radiative-shock code. MoCHII adds gas physics to the validated

MoCafe v2.00 dust radiative-transfer engine. Its founding design treats grid traversal, radiation estimators, dust physics, parallelism, file formats, and synthetic observations as first-class components. M^3 adds three-dimensional Monte Carlo radiation transport to MAPPINGS V, whose principal strength is its extensive atomic, ionic, heating, cooling, and emission-line physics [1, 3]. MAPPINGS V 5.2.1 continues that parent line as version 5.2.1, with separate photoionization, shock, equilibrium-cooling, non-equilibrium-cooling, and single-ion diagnostic drivers.

This memo distinguishes three levels of evidence:

1. *Physics present in the parent code.* The MAPPINGS V source tree contains one-dimensional photoionization, shock, non-equilibrium, dust, and plasma utilities.
2. *Physics connected to the three-dimensional driver.* A routine appearing in the source tree is not automatically active in the `montph7` Monte Carlo path.
3. *Physics quantitatively gated in the current code.* This is the strongest evidence: an analytic test, independent solver, database comparison, or published benchmark with a stated numerical tolerance.

This distinction is essential for shocks and time dependence. The public MAPPINGS tree contains `shock2--shock5`, `timion`, and `timtqui`; however, the three-dimensional `montph7` family sets `nmod='EQUI'` before the cell thermal solve. Thus the present three-dimensional M^3 calculation is an equilibrium photoionization model. The legacy routines remain valuable MAPPINGS capabilities, but they should not be described as coupled three-dimensional M^3 physics without additional development and validation.

No direct same-input, same-atomic-data timing or line-flux campaign among the three local executables has yet been performed. Numerical performance comparisons in this memo therefore use algorithm structure and each code's own recorded gates. They are diagnostic rather than a substitute for a controlled cross-code experiment.

2. Lineage: why MAPPINGS V 5.2.1 is not M^3 5.2.1

The three code names can easily obscure the most important architectural fact. M^3 is not a three-dimensional front end to the newly added MAPPINGS V 5.2.1 directory. Its source identifies itself as MAPPINGS V 5.1.13 with a separate Monte Carlo development history. The two trees still share 78 Fortran filenames, but the shared implementations have diverged substantially; the 5.2.1 versions of core files such as `photo6.f`, `photo7.f`, `shock5.f`, `mapinit.f`, and `cool.f` contain extensive later changes. M^3 adds its own `montph7`, Cartesian packet transport, PDF, and three-dimensional thermal-solver files instead [1, 4].

This gives each branch a different meaning:

- MAPPINGS V 5.2.1 is the newer parent-physics branch. Its public `map52` executable offers P6/P7 multizone photoionization, S5 shock plus precursor models, CIE and NEQ cooling curves, PIE curves, single slabs, and multilevel-ion diagnostic modes.
- M^3 is the three-dimensional geometry branch. It is the relevant comparison for arbitrary density fields, but its physics inventory must be assessed from the connected `montph7` path, not inferred from the presence of a similarly named MAPPINGS routine.

- MoCHII is an independent three-dimensional transport branch built from MoCafe rather than MAPPINGS. Similar physical terms therefore have independent algorithms, atomic-data pipelines, and validation histories.

Consequently, a discrepancy between MoCHII and M^3 cannot safely be labeled a discrepancy with MAPPINGS V 5.2.1 5.2.1. For future differential tests, 5.2.1 should be the primary one-zone and 1D microphysics reference; M^3 should be treated as the older MAPPINGS-derived 3D realization.

3. MAPPINGS V 5.2.1 in its own right

3.1 Deterministic 1D integration and diffuse radiation

P6 and P7 integrate successive zones in spherical or plane-parallel geometry. Step sizes respond to optical depth and predicted changes in temperature and ionization. The result has no Monte Carlo shot noise and is therefore efficient for large parameter grids, detailed spectra, and thin cooling zones. Isobaric models can include radiation pressure, $P(r) = P_0 + P_{\rm rad}(r)$, while other choices provide constant density or a prescribed density law. The model may terminate at a distance, an ion or hydrogen column, or an optical/radiation-bounded condition.

The price is geometric closure. The diffuse field is explicitly labeled "OUTWARD ONLY" in `photo6.f` and `photo7.f`. It follows the allowed 1D directions rather than transporting recombination photons over arbitrary solid angle. This is fast and often adequate for a monotonic sphere or slab, but it cannot represent porous escape, oblique shadows, clump illumination, or inward/backward diffuse coupling as a true 3D transport calculation can. The word "adaptive" in the S5 menu refers to an adaptive one-dimensional shock integration, not to a 3D AMR mesh.

3.2 Time dependence, cooling flows, and shocks

Unlike the public M^3 3D driver, the P6/P7 path genuinely connects a finite source lifetime to the time-dependent ion solver. It supports an equilibrium solution, a finite-age ionization calculation, and a post-equilibrium source-off evolution. The implementation advances ionic populations with adaptive substeps and couples the elapsed ionization-front time to each new zone. This is a useful one-dimensional ionization-history model, although it is not radiation hydrodynamics and does not evolve an arbitrary multidimensional density field.

The S5 path is an even clearer 5.2.1 advantage. It solves a steady, plane-parallel radiative MHD shock, adapts its step to atomic, cooling, and photon-absorption timescales, and iterates the shock radiation with an automatically calculated photoionized precursor. The source description advertises shock speeds from roughly 10 to 10^4 km s^{-1}. Companion drivers compute CIE cooling, non-equilibrium cooling, and PIE curves. For one-dimensional shock spectra and cooling flows, neither current 3D code offers an equivalent connected and mature path.

3.3 Atomic, electron, and dust physics

The standard preference file activates 16 elements, while `mapFull.prefs` activates all 30 elements from H through Zn; static dimensions permit 31 ion stages and 10,240 photon-energy entries. The thermal loop directly calls fine-structure, intercombination, resonance, multilevel, iron, recombination, free-free, collisional-ionization, charge-exchange, grain/PAH, and cosmic-ray terms. Optional κ electron distributions modify excitation, ionization, and recombination rates. These are major strengths for broad plasma diagnostics and for gas near or above coolant critical densities.

The 1D dust path is materially more complete than the public M^3 3D dust branch. In addition to grain charge, depletion, photoelectric heating, gas-grain collisions, PAHs, and destruction prescriptions, P6/P7 call `dusttemp` zone by zone. That routine implements a Draine–Li stochastic temperature distribution and returns an infrared photon field. Thus MAPPINGS V 5.2.1 has a coupled 1D gas–dust–IR calculation, whereas MoCHII has the stronger spatially resolved 3D dust transport and imaging chain.

Several cautions remain. The atomic tables are broad but globally coupled and are harder to provenance or replace ion by ion than the MoCHII registry. A Compton routine exists but no call from the examined main cooling path was found, and the microturbulent heating block in `cool.f` is commented out; neither should be counted as active 5.2.1 physics without a dedicated execution test. Resonance-line attenuation uses escape/transfer approximations rather than a multidimensional, frequency-redistributing Monte Carlo line solver.

3.4 Software and workflow limitations

MAPPINGS V 5.2.1 preserves a productive but legacy Fortran architecture: common blocks, compile-time maxima, runtime table loading, interactive menus, and many ASCII output products. P7 exposes experimental routines and an API, but the normal workflow remains less scriptable and less modular than MoCHII's namelist, HDF5/FITS, module, and Python-reader design. The documentation overview describes the manual as work in progress, so reliable use still requires source reading. The benefit of this conservatism is continuity with decades of MAPPINGS models; the cost is a higher barrier to automated regression testing, embedding, and isolated replacement of atomic data.

4. Executive comparison

Table 1: High-level comparison of the current three-dimensional codes.

Topic	MoCHII	M^3
Grid	Octree AMR or a memory-minimal uniform Cartesian backend; optional solution-driven I-front refinement.	Uniform Cartesian grid divided into equal MPI blocks; no AMR.
Absorption transport	Analytic leaf-by-leaf attenuation estimator when the band has no scattering; only direction and frequency are sampled.	Lucy-style random optical depth, discrete absorption, and immediate local re-emission.
Diffuse field	Explicit selected H/He recombination channels rebuilt from the current state; case B OTS remains available.	Local continuum and line emissivity PDF sampled immediately after gas absorption; isotropic re-emission.

Table 1 continued.

Topic	MoCHII	M^3
Frequency treatment	Dynamically allocated FUV and EUV segments, normally tens of rate-optimized bins with threshold alignment.	Up to 10,240 bins; a fixed number of packets is launched at every frequency and packet energy follows the source spectrum.
Parallel model	Packet decomposition and shared-memory grid arrays; one full grid copy per node.	Spatial domain decomposition; packets crossing blocks are synchronized through MPI.
Gas microphysics	Modern, provenance-bearing fitted data for an H II-region/PDR-centered registry; low-density cooling fits in the thermal loop.	Broad MAPPINGS ion network and multilevel line cooling inside the thermal loop; older but extensively exercised atomic-data compilation.
Dust	EUV/FUV absorption and HG scattering, live dust/PAH survival, equilibrium temperature, SEDust stochastic emission, and IR maps.	Rich grain charge, depletion, photoelectric, and PAH microphysics; the public 3D dust transport branch treats dust extinction/absorption but does not expose an equivalent scattering plus stochastic-IR imaging chain.
Synthetic observations	Leaf emissivities, flux-conserving line and field maps, dust-band maps, multi-observer peel-off images, and spectral cubes.	Extensive line spectrum and cell-resolved emissivities; maps can be made by projection, but no equivalent general observer peel-off layer is evident in the public path.
Best current use	Dusty, clumpy, spatially resolved H II regions and atomic PDR skins on simulation-derived density fields.	Broad nebular line diagnostics, denser gas, harder PN-like spectra, and problems that benefit from the full MAPPINGS ion and cooling inventory.

The concise verdict is:

> MoCHII is presently the stronger three-dimensional transport, dust, AMR, and synthetic-observation framework. M^3 is presently the stronger broad-spectrum nebular microphysics framework, especially when many ion stages or density-dependent line cooling determine the solution.

Table 2: What MAPPINGS V 5.2.1 adds to the original two-code comparison.

Topic	MAPPINGS V 5.2.1 strength	Limitation relative to the 3D codes
Photoionization	Noise-free adaptive 1D integration; equilibrium, finite-age, and source-off evolution; radiation-pressure options.	Only spherical or plane-parallel structure; outward-only diffuse field.
Shocks and cooling flows	Connected steady radiative MHD shock plus iterated precursor, CIE cooling, NEQ cooling, and PIE modes.	Steady or prescribed 1D histories, not multidimensional radiation hydrodynamics.
Microphysics	Full 30-element preference set, many ion stages, density-dependent line cooling, κ electrons, cosmic rays, and broad spectral output.	Global tables are harder to audit and update ion by ion; not every dormant routine is connected to the main cooling path.
Dust	Grain/PAH charge and destruction, gas–grain exchange, stochastic temperature distribution, and a coupled 1D IR field.	No arbitrary 3D dust geometry, shadowing, AMR, or observer image construction.
Numerics and workflow	Fast deterministic parameter grids with decades of model heritage.	Legacy interactive common-block architecture, compile-time maxima, fragmented ASCII products, and incomplete manuals.
Best current use	Integrated spectra of 1D H II regions, PNe, AGN slabs, radiative shocks, and equilibrium/non-equilibrium cooling sequences.	Not a replacement for 3D modeling when geometry is part of the physical question.

5. Radiative-transfer algorithms

5.1 Radiation-field estimators

Both three-dimensional codes estimate the cell radiation field from packet trajectories, but their estimators differ fundamentally. M^3 follows the Lucy path length estimator. At frequency ν, a packet with energy ϵ_ν crossing a distance l in cell volume V contributes schematically

$$J_\nu = \frac{1}{4\pi \Delta t V} \sum_{\text{paths}} \epsilon_\nu l. \tag{1}$$

The distance to the next absorption is sampled from

$$\tau_p = -\ln p, \tag{2}$$

with p uniform on $(0, 1)$. The packet crosses Cartesian cells until its accumulated optical depth reaches τ_p, is absorbed, and is re-emitted in situ from the local emissivity distribution. This is conceptually general and treats diffuse radiation naturally.

For the absorption-only ionizing band, MoCHII instead integrates the attenuated packet weight analytically through every crossed leaf:

$$\Delta J_b(\text{leaf}) = wL_{\text{pkt}} \frac{e^{-\tau_{\text{in}}} - e^{-\tau_{\text{out}}}}{\kappa_b}. \tag{3}$$

The packet is followed to the domain edge and contributes the expectation value of the absorbed path in each leaf. There is no sampled absorption location; statistical noise comes only from direction and bin sampling. The analytic attenuation gate measured a mean photo-rate bias of +0.004%. An optional Owen-scrambled Sobol launch (`launch_sequence='sobol'`) stratifies exactly those remaining launch variables: on the same gate the cell-wise photo-rate error then scales as $\sim 1/N$ rather than $1/\sqrt{N}$, with measured effective photon gains of 29$\times$ at 2^{17} packets and 222$\times$ at 2^{20}, and the launch set becomes independent of the MPI task count. When dust scattering is enabled, MoCHII changes to a forced-first-interaction estimator, applies the dust albedo as a packet weight, samples an HG scattering direction, and uses Russian roulette at low weight.

Equation (3) is a substantial advantage for the present MoCHII target regime: ionizing radiation in H II regions is normally absorption-dominated. M^3 pays the variance of discrete absorption events even in that limit. Conversely, the M^3 packet cycle is more general for a diffuse spectrum containing many continuum and line channels, whereas the current MoCHII diffuse emitter is explicitly limited to selected recombination channels.

MAPPINGS V 5.2.1 has neither Monte Carlo estimator. Its zone integrator is deterministic and therefore essentially noise-free, but achieves that efficiency by imposing spherical or plane-parallel symmetry and an outward-only closure for the diffuse field.

5.2 Diffuse radiation and energy packets

In M^3, gas absorption converts the stellar packet into a diffuse packet immediately. Its new frequency is drawn from a local emissivity PDF containing recombination continua and lines, and its new direction is isotropic. This implements a full three-dimensional alternative to outward-only diffuse-field approximations [1].

MoCHII offers two controlled limits. Case B uses on-the-spot recombination. With the explicit diffuse field enabled, case A is used and the emission table is rebuilt each iteration from the current state. The principal channels are H II, He II, and He III ground recombination continua; the optional He I excited channel adds the 19.82-eV, 584-Å, and two-photon contributions. These packets use the same transport estimator as stellar packets. This split makes case-A/case-B bracketing and energy accounting transparent, but it is not yet the full local emissivity inventory sampled by M^3.

5.3 Spectral discretization

The public M^3 compile-time ceiling is 10,240 frequency entries. The Monte Carlo loop launches N packets for every frequency interval; packet energy is proportional to $L_\nu \Delta\nu/N$. This stratified sampling guarantees representation of weak spectral intervals and is attractive for detailed continua. Its cost scales approximately with $N_\nu N$ rather than with one total packet budget, and its large $J_\nu(x, y, z)$ and opacity arrays are major memory consumers.

MoCHII samples a frequency bin from the luminosity CDF and assigns every packet the same band luminosity. Its bins are designed around rate integrals rather than spectral oversampling. The ionizing grid can align edges with H, He, and active metal thresholds, while an independent FUV segment preserves the EUV resolution. This is efficient for ionization and thermal balance, but a fine line-rich diffuse spectrum would require many more bins and packets than the normal 32–64-bin production setup.

5.4 Velocity fields and line transport

M^3 reads a three-dimensional velocity field and applies a projected Doppler shift to the packet frequency bin during continuum transport. MoCHII currently has no velocity-dependent band transport. This is a real M^3 advantage for bulk-flow effects on continuum opacity.

None of the three codes should yet be treated as a general resonance-line transfer solver. M^3 includes resonance emissivities but, in the published implementation, assigns resonance lines the same cross section as the continuum. This is not a frequency-redistributing resonant-scattering calculation. MoCHII explicitly treats its diagnostic lines as optically thin emissivities and leaves resonance transfer to a dedicated code such as LaRT. MAPPINGS V 5.2.1 applies geometry-dependent escape and attenuation approximations rather than multidimensional redistribution. For forbidden and recombination lines this is normally adequate; for Lyα, C IV, Mg II, or line profiles in moving media, neither current path is sufficient.

6. Grid algorithms, resolution, and parallelism

6.1 Octree AMR versus a uniform Cartesian mesh

The primary spatial difference is adaptive resolution. M^3 accepts an arbitrary density distribution on a Cartesian mesh, but every cell at a given global resolution exists. The published HII20, HII40, and PN150 tests used 33^3 and 55^3 grids. Most integrated lines agreed within 10%, but I-front and boundary tracers such as [O I], [O II], and [S II] could differ by more than 10% [1]. This is the expected weakness of a uniform grid: the thin zone that most needs resolution occupies a small part of the volume.

MoCHII can ingest a pre-existing octree or rebuild the octree after an initial solution. A leaf is refined when it lies in the I-front or sees a large face-neighbor ionization jump. In the recorded gate, refinement from a level-5 start to level 7 produced 464,248 leaves instead of the 2,097,152 cells of a complete level-7 grid; the effective ionized radius agreed with the native level-7 result by 0.008%. The refinement follows clump skins and distributed fronts in turbulent media, not merely the source position.

For problems that truly require a uniform mesh, MoCHII also has a raster-indexed Cartesian backend. It allocates no octree topology, neighbor table, center array, or leaf maps. The dedicated incremental Amanatides–Woo DDA precomputes $t_{\max}$ and $t_{\triangle}$ per ray and advances by comparisons and additions. In the recorded Strömgren test it was twice as fast as the octree traversal, while matching the solution to rounding. At 512^3, omitting the tree and geometry arrays saves approximately 10 GB before physical fields are counted.

6.2 MPI decomposition and scaling trade-offs

M^3 spatially decomposes the Cartesian domain into equal blocks. Each rank stores the local thermal and ionization state, and packets crossing a block boundary are transferred through collective MPI operations. Spatial decomposition is attractive when no node can hold the full grid. It also introduces frequent synchronization: the current transport implementation collectively reduces large packet-state arrays containing flags, positions, directions, frequencies, and absorption counts. At large N_ν, packet count, and MPI task count, communication can dominate.

MOCHII decomposes packets rather than space. The octree and leaf fields reside in MPI-3 shared memory, so ranks on one node share one copy; each node nevertheless holds a complete grid. Ranks transport independent packets and reduce the accumulated radiation tally at the end of a pass. This minimizes packet communication and is well matched to Monte Carlo transport, but the requirement of a full grid on each node ultimately limits the largest possible AMR problem. Thus the scaling comparison is not one-sided: MOCHII should normally scale more cleanly in packet number, whereas M^3 can distribute spatial memory more aggressively. MAPPINGS V 5.2.1 avoids the 3D memory problem entirely by advancing one 1D structure, which is a decisive throughput advantage only when its symmetry assumptions are physically defensible.

7. Ionization, thermal balance, and atomic data

7.1 Element and ion inventories

The local M^3 default atom file activates 16 elements: H, He, C, N, O, Ne, Na, Mg, Al, Si, S, Cl, Ar, Ca, Fe, and Ni. Static array dimensions allow up to 30 elements and 31 ion stages. The published description characterizes the parent MAPPINGS V system as containing 30 elements and roughly 80,000 cooling and recombination lines [1]. The distinction between the parent capacity and the default public data set should be retained when quoting these numbers. The new MAPPINGS V 5.2.1 audit resolves the parent-code side directly: `mapStd.prefs` is the 16-element setup and `mapFull.prefs` activates 30 elements from H through Zn. This full preference set, rather than the M^3 default, is the relevant breadth reference.

MOCHII currently treats H and He plus an eleven-element metal registry: C, N, O, Ne, S, Ar, Mg, Fe, Si, Cl, and Ca. The tracked stages are chosen for H II regions and atomic PDR skins, rather than extending to the highest charge state of every nucleus. New ions are added through provenance-bearing element, cooling-fit, and *n*-level files. This is much easier to audit and extend than the global MAPPINGS tables, but the present inventory is less suitable for hard PN, X-ray, or coronal-ion problems.

7.2 Photoionization, recombination, and charge exchange

MOCHII uses exact hydrogenic ground-state cross sections for H I and He II, and Verner et al. [6] fits for He I and metals. Hydrogen and helium recombination defaults to the Badnell (2023) radiative total minus the Mao–Kaastra (2016) ground-state coefficient (case B), with the Hui–Gnedin fits retained as an option; metal RR and DR are derived primarily from Badnell/CHIANTI v11.0.2 data. Charge exchange uses the modern first-stage rates of Huang et al. and the higher-stage Kingdon–Ferland compilation. Each generated file records its source, fit range, formula, and maximum fit error.

M^3 uses the established MAPPINGS compilation. The published code description identifies photoionization data from Seaton, Raymond, Osterbrock, and Gould, including Auger corrections, and separates hydrogenic from non-hydrogenic recombination. Much of this compilation is older than the MOCHII sources, but it spans more ions and has been exercised in many MAPPINGS applications. “Newer” is not automatically equivalent to “more accurate”: line predictions can differ by several to tens of percent among reputable collision-strength and A-value compilations, especially for iron. The decisive test is therefore ion-by-ion comparison with direct database evaluation and observation, not vintage alone.

7.3 Electron distributions and secondary processes

Both M^3 and MAPPINGS V 5.2.1 support either a Maxwellian or a κ electron energy distribution, with rate corrections supplied by KAPPA data. This is useful when exploring non-Maxwellian interpretations of nebular diagnostics. MOCHII assumes a Maxwellian distribution.

All three codes include secondary-ionization concepts. MOCHII optionally partitions the energy of photoelectrons above 40 eV among heat and secondary H/He ionizations following Shull–van Steenberg. The MAPPINGS ion-balance routines call a secondary-ionization module and also include cosmic-ray ionization/heating. The MAPPINGS-derived codes therefore have the broader general-plasma framework. A Compton routine is present in both source trees, but no call from either examined main cooling path was found; it should not be counted as active physics without a dedicated test.

7.4 The critical difference: density-dependent metal cooling

This is the most important gas-physics advantage of both MAPPINGS-derived codes over MOCHII. Their thermal solvers call fine-structure, intercombination, multilevel, and iron line modules during every evaluation of the net cooling function. Level populations and collisional de-excitation therefore affect the equilibrium temperature when n_e approaches or exceeds a transition’s critical density.

MOCHII deliberately separates two products:

- analytic Tier-1 cooling fits, evaluated cheaply inside the thermal bisection in the optically thin low-density limit; and
- Tier-2 n-level statistical-equilibrium data, evaluated on the converged state to produce density-dependent diagnostic emissivities.

The output line ratios therefore respond to density, but collisional quenching does not feed back into the hot thermal loop. At ordinary H II-region densities this is often acceptable. In compact H II regions, dense PN, shocked cooling zones, or any model with $n_e \gtrsim n_{crit}$ for a dominant coolant, M^3 is physically better founded. Moving the generic MOCHII n-level solver or a reduced density-suppression fit into the thermal loop is the highest-value gas-physics improvement suggested by this comparison.

7.5 Thermal processes and equilibrium solution

All three codes solve ionization and temperature under the current radiation field, iterating the spatial solution where required. MoCHII brackets the root of heating minus cooling in $\log T_e$ and re-solves the H/He and metal ionization at every trial temperature. Its current budget includes photoionization heating, H/He recombination and collisional losses, free-free emission with a tabulated Gaunt integral (H/He and, by default, the trace metals with their net ion charge), metal-line cooling, optional metal photoheating, secondary ionization, and the atomic-PDR grain and neutral collision terms.

The MAPPINGS thermal solver used by MAPPINGS V 5.2.1 and inherited by M^3 includes a broader traditional plasma budget: hydrogen and helium line cooling, fine/intercombination/resonance and multilevel metal cooling, free-free, collisional ionization, recombination, photoionization heating, charge-exchange energy exchange, cosmic rays, grain/PAH photoelectric terms, and grain collisional cooling. The microturbulent dissipation block in the examined 5.2.1 `cool.f` is commented out and is therefore not listed as active. This breadth and the direct multilevel coupling are major MAPPINGS strengths.

8. Dust, PAHs, and PDR physics

8.1 Where the MAPPINGS microphysics is stronger

The MAPPINGS grain microphysics is detailed. The local source tree carries graphite and silicate size distributions, depletion bookkeeping, grain charge balance, electron and ion sticking, grain photoelectric heating, collisional grain cooling, PAH charge states, and optional thermal or ionization-parameter-dependent grain destruction. For the local exchange of energy and charge between gas and a resolved grain population, this is more microscopic than the current MoCHII Bakes–Tielens heating prescription.

In MAPPINGS V 5.2.1, this layer is connected to the P6/P7 zone integration and to `dusttemp`, which computes a Draine–Li stochastic temperature distribution and an IR photon field. It is therefore a complete 1D dust emission path, not merely a library capability. The distinction below is specific to the public M^3 three-dimensional branch.

8.2 Where the three-dimensional M^3 dust path is limited

The public `montph7_dust` branch includes dust in the extinction and terminates a packet when the selected interaction is grain absorption. The dust data contain absorption, scattering, and asymmetry information, but the examined 3D branch computes an absorption fraction and does not sample a new dust-scattering direction. The dedicated `dusttemp` routine is called from the one-dimensional `photo6/7` family, not from the public 3D driver. Consequently, the rich MAPPINGS grain microphysics should not be conflated with a complete 3D dust radiative transfer and emergent-IR calculation.

This does not imply that dust is absent from M^3: dust opacity, depletion, grain heating/cooling, and PAH processes affect the gas. It means that the public path does not presently expose a validated chain equivalent to MoCHII's dust scattering, absorbed-energy ledger, stochastic emission, and observer maps.

8.3 The MoCHII dust chain

MoCHII couples dust to the same FUV/EUV radiation field that sets the gas state:

1. Gas and dust compete for ionizing photons using bin-dependent absorption cross sections. The dusty Strömgren gate reproduces the one-dimensional radius to +0.69% and the classic radius reduction $R/R_0 = 0.735$.
2. With scattering enabled, the extinction includes the dust scattering coefficient, a forced first event is sampled, and the packet follows an HG direction with an albedo weight. The dusty-sphere result lies between the pure-absorption and extinction-as-absorption bounds.
3. The live dust model updates the dust abundance from the computed neutral fraction. A separate PAH survival fraction permits destruction inside ionized gas and produces a PAH-bright I-front ring.
4. The absorbed FUV/EUV power gives a leaf equilibrium mixture temperature and an energy-conserving IR spectrum.
5. SEDust supplies astrodust, DL07, or Zubko stochastic emission. The final normalization enforces $\int L_\lambda \, d\lambda = \sum_{\rm leaf} H_{\rm dust} V$; the smoke test closes at unity and resolves the usual PAH bands.
6. Leaf SEDs at selected wavelengths generate spatially resolved 8, 24, 70, and 160-μm maps.

For dusty three-dimensional H II regions, this end-to-end chain is the clearest overall physical advantage of MoCHII.

8.4 PDR scope

MoCHII extends the band below 13.6 eV so that low-threshold metals and dust see the FUV field beyond the H I-front. Optional switches include electrons from the metal cascade, metal photoheating, Bakes–Tielens grain photoelectric heating plus recombination cooling, and neutral-H impact cooling in [C II] 158 μm and [O I] 63 μm. This is an atomic PDR skin, not a complete PDR code. H_2/CO chemistry, molecular self-shielding, molecular cooling, line transfer, and gas–dust collisional coupling remain outside the model.

The MAPPINGS trees likewise contain more grain and general plasma physics than a minimal H II solver, but the present comparison did not identify a full molecular H_2/CO network in either the 5.2.1 photoionization path or the public M^3 3D driver. None of the three codes should be selected as a substitute for a dedicated molecular PDR code without additional chemistry.

9. Emission lines, continua, and synthetic observations

9.1 Line inventories and emissivity accuracy

The MAPPINGS lineage was designed to predict a very broad line spectrum. The M^3 cooling and output path includes resonance, intercombination, forbidden, fine-structure, hydrogenic, helium, heavy recombination, and extensive iron-line data. Its five-level and multilevel populations are already part of the thermal solution. This makes M^3 attractive for integrated diagnostic grids and hard nebulae for which a large fraction of the periodic table and many ion stages matter.

MAPPINGS V 5.2.1 is the strongest version of that argument when 1D geometry is acceptable: the full preference set covers 30 elements, and the deterministic zone solver can afford a detailed spectrum without Monte Carlo sampling noise. M^3 trades some parent-code recency and 1D special modes for arbitrary Cartesian geometry.

MoCHII has a narrower but highly auditable diagnostic layer. Every registered ion with a Tier-2 file is solved by the same partial-pivot statistical-equilibrium code using fitted CHIANTI v11 effective collision strengths and explicit A-values. H I and He II recombination lines use Storey–Hummer case-A/B tables. The nebular continuum includes free-bound plus free-free and two-photon components. On an identical state, the Fortran line luminosities match the independent Python solver to at least four significant figures. The limitation is coverage, not the internal emissivity algebra.

9.2 Images and observing geometry

MoCHII writes leaf state and line-emissivity blocks to HDF5/FITS. Its Python reader deposits each AMR leaf with exact pixel-area overlap, so surface-brightness integration recovers the three-dimensional luminosity to machine precision. It also supports slab cuts and EM-, electron-, hydrogen-, or volume-weighted field maps.

For the transported FUV/EUV field, an extra converged-state pass applies observer peel-off. Stellar and diffuse emission contribute direct $e^{-\tau}$ terms, every dust interaction contributes an HG-weighted scattered term, and the optical-depth walk uses the same gas and dust opacity as the rate calculation. Multiple observers, unattenuated direct images, and bin-resolved cubes are available. The direct thin-medium gate matches the analytic image-integrated flux by +0.15% under the current defaults (exact hydrogenic cross sections, threshold-aligned bins).

M^3 outputs cell-resolved physical quantities and line emissivities, and its published applications construct spatial line maps from them. This is adequate for optically thin line projection. The public driver does not show a comparable general next-event estimator for arbitrary observers, nor the separation of direct and dust-scattered continua. Thus M^3 is richer in the number of intrinsic lines, while MoCHII is richer in the chain that turns a three-dimensional state into calibrated synthetic observations.

10. Convergence, verification, and maturity

10.1 Convergence criteria

M^3 updates the radiation field, solves each illuminated cell, and counts cells whose temperature and H$^+$ changes meet the selected tolerances. The run stops when a requested fraction of ionized cells has converged or after the iteration limit. Its public defaults use percent- level cell changes and increase the packet count when convergence stalls. This is intuitive, but the result is sensitive to how the ionized-cell mask and required fraction are chosen.

MoCHII reports both cell-max and volume-integrated pairs:

$$\delta_x^{(V)} = \frac{\sum V\,|\Delta x_{\rm HII}|}{\sum V x_{\rm HII}}, \qquad \delta_T^{(V)} = \frac{\sum V n_{\rm e}\,|\Delta T_{\rm e}|}{\sum V n_{\rm e} T_{\rm e}}. \tag{4}$$

The volume criterion suppresses front-cell jitter by the small volume of the front and excludes temperature oscillations in nearly electron-free cells. In the FUV dusty test, cell maxima remained at 0.1 and 6.4 while the volume measures reached stable Monte Carlo floors of 2.7×10^{-3} and 6.6×10^{-3}. Thirty further iterations changed the ionized volume by only 0.014%. This is a stronger convergence diagnostic for turbulent, clumpy grids, provided the tolerance is set just above the packet-noise floor.

10.2 Validation evidence

The MAPPINGS lineage has the stronger external maturity record. M^3 inherits decades of MAPPINGS development and has been published against the HII20, HII40, and PN150 Lexington/Meudon standards. Its three-dimensional blister, bipolar, and fractal applications demonstrate the importance of geometry for spatially resolved line diagnostics [1, 2]. MoCHII is newer, but its current repository contains unusually granular gates:

- analytic H/He attenuation and rate integrals;
- a Strömgren sphere against an exact one-dimensional reference;
- AMR versus uniform and solution-driven refinement versus native high resolution;
- H/He thermal balance against an independent Gauss–Seidel reference;
- direct CHIANTI fit evaluation and PyNeb line-ratio checks;
- MOCASSIN HII20/HII40 state and line comparisons;
- case-A/case-B diffuse-field bracketing;
- dust absorption/scattering brackets and IR energy closure;
- direct peel-off flux against an analytic observer solution; and
- exact three-dimensional emissivity-to-map luminosity conservation.

These gates provide strong internal evidence and make regressions easy to localize. They do not yet replace the independent user base and long publication history of MAPPINGS. The correct conclusion is that the MAPPINGS lineage is more mature externally, while MoCHII is more explicit and fine-grained in its current numerical audit trail. MAPPINGS V 5.2.1 does not have a Monte Carlo convergence problem: zone step control and the global shock/precursor iteration dominate instead. Its decades of use and broad model suite are a major maturity advantage, but the inspected 5.2.1 tree does not provide a gate hierarchy as explicit as the current MoCHII test directories. A controlled local benchmark is still required before attributing any line difference to geometry rather than to the 5.1.13-versus-5.2.1 atomic and solver divergence.

11. Software architecture and maintainability

MoCHII uses Fortran modules, allocatable leaf arrays, a single namelist, MPI-3 shared memory, HDF5/FITS sections, and a data-driven species registry. New atomic products are generated offline by Python tools and carry machine-readable

provenance. A new ion normally requires no new solver code. The transport, ion balance, thermal balance, lines, dust, and output layers are separated modules. This architecture substantially reduces the cost and risk of adding physics.
The public M^3 tree retains the Fortran 77 MAPPINGS architecture: large common blocks, compile-time maxima, interactive input, and many global arrays. Several historical `montph7_*` variants duplicate large parts of the transport driver. This structure has the advantage of preserving a mature code base and its full menu of MAPPINGS models, but it makes a change to the 3D algorithm harder to propagate and test. The available Sphinx input/output documentation is sparse, so reliable use often requires reading the source and reproducing the authors' setup.
The architectural verdict is therefore clear: MoCHII is easier to extend, audit, automate, and connect to simulation data. M^3's value lies primarily in the depth of the inherited physical library, not in modern software structure.
MAPPINGS V 5.2.1 shares the same common-block and interactive heritage, but its organization is cleaner than the collection of duplicated M^3 `montph7_*` experiments and it has a single public `map52` model selector. Runtime preference files make the 16- or 30-element atom sets configurable without recompilation. Nevertheless, automated model grids require scripting interactive input and parsing many ASCII outputs; the current documentation overview explicitly remains incomplete. For maintainability the ordering is MoCHII first, MAPPINGS V 5.2.1 second, and the experimental M^3 3D fork third.

12. Suitability by science case

Table 3: Recommended code by present science requirement.

Science requirement	Preferred code	Reason
Clumpy or porous H II region from a TNG/RAMSES snapshot	MoCHII	AMR, front refinement, and simulation-oriented I/O.
Ionization-front position, shadows, and clump skins	MoCHII	Adaptive resolution and the analytic absorption estimator.
Dusty Strömgren structure and UV attenuation	MoCHII	Gas–dust competition and validated HG scattering in the 3D band.
PAH rings, IR SEDs, or 8–160-μm maps	MoCHII	SEDust, leaf dust emissivities, and exact absorbed-energy normalization.
Spatially resolved IFU line maps	Usually MoCHII	AMR emissivities and flux-conserving projection; verify that the required ions exist.
Very broad integrated optical/UV/IR line inventory	MAPPINGS V 5.2.1	Current full 30-element MAPPINGS preference set and deterministic detailed spectrum.

Table 3 continued.

Science requirement	Preferred code	Reason
Compact high-density nebula	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	Both feed density-dependent multilevel cooling back into T_e; geometry decides.
Hard PN-like ionizing spectrum	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	More high charge states; M^3 additionally has published PN150 3D validation.
Non-Maxwellian κ electron experiment	MAPPINGS V 5.2.1 or M^3	Both MAPPINGS-derived trees have native κ rate support.
One-dimensional radiation-pressure-confined nebula	MAPPINGS V 5.2.1	Isobaric photoionization can include the accumulated radiation pressure.
Finite source lifetime or source-off ionization history in 1D	MAPPINGS V 5.2.1	P6/P7 connect finite-age modes to the time-dependent ion solver.
Steady 1D radiative MHD shock and precursor	MAPPINGS V 5.2.1	S5 couples adaptive post-shock cooling to an iterated auto-preionization solution.
CIE, NEQ, or PIE cooling-curve grid	MAPPINGS V 5.2.1	Dedicated CC, NC, and PP drivers and a broad general-plasma ion inventory.
Controlled test of modern CHIANTI v11 coefficients	MoCHII	Versioned fitted data and direct fit-error gates.
Three-dimensional time-dependent ionization	None	Both public 3D drivers are equilibrium solvers; MAPPINGS V 5.2.1 time dependence is 1D.
Three-dimensional radiative shock	None	MAPPINGS V 5.2.1 has a strong 1D shock mode, but it is not coupled to either 3D transport architecture.
Molecular H_2/CO PDR	None	All three need a dedicated molecular chemistry and line-transfer layer.
Lyα or metal resonance-line profiles	None	Requires frequency-redistributing resonant transfer, e.g. LaRT.

13. Recommended development path for MoCHII

The comparison suggests a focused roadmap rather than wholesale replacement of either code.

1. **Put density dependence into the thermal cooling.** Reuse the generic n-level solver in the thermal loop for dominant coolants, or fit a suppression factor versus (T_e, n_e) around each critical density. This closes the most consequential gap relative to MAPPINGS.
2. **Extend the registry upward in ion stage.** Add the ions needed for PN and hard-field applications before adding marginal new low stages. The existing data-operation design is well suited to this.
3. **Add Na, Al, and Ni only when driven by a diagnostic need.** Their MAPPINGS implementations provide useful reference behavior, while the MoCHII pipeline should retain explicit provenance and ion-by-ion gates.
4. **Retain the present transport architecture.** The octree, analytic estimator, Cartesian DDA, threshold-aligned bins, and peel-off layer are major advantages and should not be replaced by the legacy M^3 packet synchronization model.
5. **Use MAPPINGS V 5.2.1 as the differential microphysics oracle.** Construct one-zone tests at fixed (T_e, n_e, J_ν) and compare ion fractions, heating/cooling terms, and individual line emissivities before comparing full 3D nebulae. Use M^3 only as the second-stage test of how that inherited microphysics behaves in arbitrary geometry.
6. **Perform a controlled common-physics benchmark.** Use the same uniform grid, source SED, abundances, dust-off setting, frequency range, and atomic subset. Compare J_ν, photo-rates, ion fractions, T_e, cooling budgets, and line emissivities separately. Only then compare total runtime and MPI scaling.

The most valuable hybrid would combine MAPPINGS V 5.2.1-like density-dependent microphysics with the current MoCHII geometry, transport, dust, and observer layers. This would preserve the strongest feature of each code without inheriting the principal numerical and software limitations of the other.

14. Final assessment

Capability	Stronger now	Confidence
Three-dimensional transport estimator	MoCHII	High
Adaptive spatial resolution	MoCHII	High
Dust scattering and IR/PAH observables	MoCHII	High
Observer imaging and data products	MoCHII	High
Atomic-data provenance and fit auditability	MoCHII	High
Element and ion-stage breadth	MAPPINGS V 5.2.1	High
Density-dependent thermal line cooling	MAPPINGS V 5.2.1/M^3	High
Hard PN/general plasma coverage	MAPPINGS V 5.2.1	High
Finite-age 1D photoionization	MAPPINGS V 5.2.1	High
Radiative MHD shock and precursor	MAPPINGS V 5.2.1	High
Noise-free 1D parameter-grid throughput	MAPPINGS V 5.2.1	High
External validation history	MAPPINGS lineage	High
Software extensibility and automation	MoCHII	High
Absolute line accuracy for every ion	Case dependent	Requires a common-data benchmark
Massively distributed memory ceiling	Architecture dependent	Requires a scaling experiment

For the principal intended application–dusty, structured, spatially resolved H II regions–MoCHII is the better current platform. For broad integrated emission-line diagnostics, dense nebulae, harder ionization conditions, finite-age 1D photoionization, and radiative shocks, MAPPINGS V 5.2.1 is the more complete physical reference. M^3 becomes preferable only when those MAPPINGS-like line and cooling strengths must be combined with arbitrary three-dimensional Cartesian geometry. None of these judgments is universal: first decide whether geometry, time history, shock dynamics, dust observables, density-dependent cooling, or ion-stage breadth is the controlling requirement.

15. Implementation evidence consulted

The principal local sources used for this audit were:

- MoCHII: `docs/PLAN.md`, `docs/MoCHII_physics.tex`, `docs/MoCHII_UserGuide.tex`, the transport/grid modules under `src/`, and the quantitative gates under `tests/`.
- M^3: `README.md`, `lab/data/ATDAT.txt`, `src/mastercode/montph7.f`, `montph7_dust.f`, `crosssections.f`, `mcteequi.f`, `teequi2.f`, `equion.f`, `cool.f`, `multilevel.f`, `grainpar.f`, `hgrains.f`, `hpahs.f`, `dusttemp.f`, `timion.f`, and `shock2--shock5.f`.

- MAPPINGS V 5.2.1: `README.md`, `src/mappings.f`, `photo6.f`, `photo7.f`, `teequi.f`, `teequi2.f`, `timion.f`, `timtqui.f`, `cool.f`, `dusttemp.f`, `shock5.f`, `neqc.f`, `coolc.f`, `transferline.f`, `lab/mapStd.prefs`, `lab/mapFull.prefs`, and `docs/DocumentionOverview.html`.

The public M^3 source contains several historical or experimental driver variants. Statements in this memo refer to the main `mastercode/montph7.f` and explicitly selected `montph7_dust.f` paths unless otherwise stated. Statements about MAPPINGS V 5.2.1 refer to the connected public 5.2.1 `map52` P6/P7/S5/CC/NC/PP paths, not merely to routines present in the directory.

References

[1] Jin, Y., Kewley, L. J., & Sutherland, R. S. 2022, "Messenger Monte Carlo MAPPINGS V (M^3)–A Self-consistent, Three-dimensional Photoionization Code," ApJ, 927, 37, https://doi.org/10.3847/1538-4357/ac48f3.

[2] Jin, Y., Kewley, L. J., & Sutherland, R. S. 2022, "Theoretically Modeling Photoionized Regions with Fractal Geometry in Three Dimensions," ApJL, 934, L1, https://doi.org/10.3847/2041-8213/ac80f3.

[3] Sutherland, R. S., & Dopita, M. A. 1993, "Cooling Functions for Low-density Astrophysical Plasmas," ApJS, 88, 253.

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[5] Lucy, L. B. 1999, "Computing Radiative Equilibria with Monte Carlo Techniques," A&A, 344, 282.

[6] Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, "Atomic Data for Astrophysics. II. New Analytic Fits for Photoionization Cross Sections of Atoms and Ions," ApJ, 465, 487.